Volume 26
Number 31
21 August 2024
Pages 20721–21306

# PCCP

Physical Chemistry Chemical Physics

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ISSN 1463-9076

ROYAL SOCIETY OF CHEMISTRY

**PAPER**
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Cite this: *Phys. Chem. Chem. Phys.*, 2024, **26**, 20750

# A first-principles alternative to empirical solvent parameters†

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Received 13th May 2024,
Accepted 24th June 2024

DOI: 10.1039/d4cp01975j

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The use of solvents is ubiquitous in chemistry. Empirical parameters, such as the Kamlet–Taft parameters and Gutmann donor/acceptor numbers, have long been used to predict and quantify the effects solvents have on chemical phenomena. Collectively however, such parameters are unsatisfactory, since each describes ultimately the same non-covalent solute–solvent and solute–solute interactions in completely disparate ways. Here we hypothesise that empirical solvent parameters are essentially proxy measures of the electrostatic terms that dominate solvent–solute interactions. On the basis of this hypothesis, we develop a new fundamental descriptor of these interactions, $\mathfrak{p}^{\pm}_{\text{solvent}}$, and show that it is a self-consistent, probe-free, first principles alternative to established empirical solvent parameters.

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## Introduction

Much of chemistry occurs in a solvent.[1–3] Solvents are used broadly in mineral processing, synthetic processes, paints, inks, personal care and cleaning products, as well as throughout the biological and chemical sciences. The biochemical processes underpinning terrestrial life itself require water. The non-covalent interactions between solutes and their solvent environment fundamentally change molecular structures,[4] solubilities,[5] properties[6] and reactivities.[7–9] Predicting and understanding the interactions between solutes and solvents, and thus the effects of solvents on equilibrium and non-equilibrium phenomena, remains one of the principal challenges facing chemists.[10]

Several empirical solvent parameters or relationships have been proposed to describe these effects (as summarised in Table 1). Generally, however, these parameters have been proposed for, and used in, specific contexts. For instance, the Kamlet–Abboud–Taft π* scale[11] of solvent polarity/polarisability has been used to rationalise the solvent's effect on Menshutkin $S_N2$ reaction rates in aprotic solvents and free energies of transfer between solvents.[12] The Kamlet–Taft α and β scales[13,14] of hydrogen bond donating and accepting capability, respectively, have been correlated with solvatochromic and nuclear magnetic resonance (NMR) shifts of fluorescent coumarin probe molecules.[15] The Hildebrand solubility parameter,[16,17] $\delta_H$, empirically accounts for free energies of solution, or transfer, *via* the energy cost of creating a cavity and disrupting solvent–solvent interactions. The Dimroth–Reichardt $E_T(30)$ values[18] (which are molar transition energies) estimate the polarity of the solvent from the position of the absorption maximum of pyridinium *N*-phenolbetaine. The Kosower *Z*-values[19] are calculated in the same manner but use 1-ethyl-4-methyoxycarbonylpyridinium iodide. The Grunwald–Winstein equation[20] was derived from the rate of *tert*-butyl chloride solvolysis in terms of the solvent's ionising power (*Y*), in 80% ethanol–solvent mixtures. This was later extended to include a second term, *lN*, representing the system's susceptibility (*l*) to the solvent's nucleophilicity (*N*).[21] Catalan *et al.*[22,23] developed scales very similar to the Kamlet–Taft parameters, however using alternate probe molecules (*i.e.* 3,20-di-*tert*-butyl-2,2,21,21-tetramethyl-5,7,9,11,13,15,17,19-docosanonaene (ttbP9)) capable of isolating polarisability from dipolarity, acidity and basicity terms and therefore overcoming some of the limitations of the π* scale.

While each of these solvent parameters has shown success in a specific purpose or context, collectively they are unsatisfactory; each seeks to describe and quantify what are ultimately the same non-covalent solute–solvent and solute–solute interactions, in disparate and empirical ways. Their formulation generally is based on a specific probe molecule (*e.g.* betaine, $SbCl_5$, $Et_3PO$, $BF_3$, ttbP9, *etc.*) or ion (*e.g.* $Na^+$, $K^+$, $Ag^+$, *etc.*), and often use an inert solvent (*e.g.* 1,2-dichloroethane) environment, meaning that the "solvent" of interest acts as a molecular solute. For instance, the Gutmann–Mayer acceptor number (AN) is a measure of a solvent's Lewis acidity and is derived from scaled $^{31}P$ NMR chemical shifts (in ppm) of varied

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† Electronic supplementary information (ESI) available. See DOI: https://doi.org/10.1039/d4cp01975j



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Table 1 Empirical solvent parameters

| Parameter | Measures | Molecular probe(s) | Comments | Parameter | Measures | Molecular probe(s) | Comments |
|---|---|---|---|---|---|---|---|
| Kamlet–Taft β-scale[24,25] | Solvent hydrogen-bond acceptor basicity | Multiple: 4-nitroaniline; *N*,*N*,-diethyl-4-nitroaniline; 4-nitrophenol; 4-nitroanisole | Values are averaged over multiple probes and make implicit assumptions about non-hydrogen bonding solvents and amphiprotic solvents. Various values are available for each solvent.[24,25] $\beta$ is also related to formation constants for H-bonding complexes and $^{19}$F NMR shifts. $\alpha$ is also related to other spectral properties, equilibria and free energies of transfer. | Kosower *Z*-Value[26] | Solvent polarity | 1-Ethyl-4-(methoxycarbonyl) pyridinium iodide | $Z = 2.859 \times 10^4\ \lambda^{-1}$, where $\lambda$ = longest-wavelength adsorption maximum of the probe. Analogous to the $E_T(30)$ scale, just with a different probe molecule. |
| Kamlet–Taft α-scale[27] | Solvent hydrogen-bond donor acidity | Multiple: 4-nitroanisole; betaine; merocyanine | | Grunwald-Winstein[28] | Solvent polarity | *t*-Butyl chloride | Correlates solvent polarity, or ionising power ($Y$), susceptibility ($l$) and nucleophilicity ($N$) with the rate of solvolysis *ks via* $\log\left(\frac{ks}{k0}\right) = mY + lN + c$ $m$ = substrate-specific parameter (assumed 1 for *t*-butyl chloride) and $k_0$ = rate of solvolysis in 80 : 20 EtOH : water v/v. |
| Kamlet–Abboud–Taft π*-scale[29] | Solvent polarity and polarisability | 7 Indicators which (i) had symmetric intense bands of maxima, (ii) $\nu_{max}$ for HBD and non-HBD solvents, (iii) minimally influenced by band overlap, (iv) adequate response to changing solvent polarity (40 for expansion) | | Gutmann–Mayer Acceptor Number[30] | Solvent acidity | $Et_3PO$ | Scaled $^{31}$P-NMR chemical shift AN = 2.21 × (δ − 41.0). AN = 0 for hexane (δ = 41.0 ppm and AN = 100 for $SbCl_5$ (δ = 86.1 ppm) |
| | | | | Gutmann Donor Number[31] | Solvent basicity | $SbCl_5$ | Enthalpy of formation (kcal mol$^{-1}$) of $SbCl_5$ in a given solvent, relative to 1,2-dichloroethane (DN for 1,2-dichloroethane is 0 by definition). |
| Hildebrand solubility parameter ($\delta_H$)[32,33] | Cavity energy | — | $\delta_H = \sqrt{-\frac{E_{vap}}{V}}$, where $E_{vap}$ is the molal heat of vaporisation and $V$ is the molal volume, is the energy required to form a cavity in a solvent to fit a solute. | Marcus Softness parameter ($\mu$)[34] | Softness/cation solvency | $Ag^+$, $Na^+$ and $K^+$ | $\mu = \frac{\Delta_{tr}G^\infty(Ag^+) - 0.5[\Delta_{tr}G^\infty(Na^+) + \Delta_{tr}G^\infty(K^+)]}{100\ \text{kJ mol}^{-1}}$ where $\Delta_{tr}G^\circ$ is the Gibbs free energy of transfer of the chosen solvent. |
| Koppel–Palm equation | Varied LSER[29] | — | Consists of distinct parameters describing non-specific solute–solvent interactions $yY$ (from permittivity, $\varepsilon$), and $pP$ (from refractive indices, $\eta D$) and specific solute–solvent interactions ($eE$, and $bB$). | Dimroth–Reichardt ($E_T(30)$) | Solvent ionising power (related to a solvent's H-bond donating ability) | Betaine 30 (Reichardt's Dye) | $E_T(30) = 2.859 \times 10^4 \lambda^{-1}$, where $\lambda$ = longest-wavelength adsorption maximum of the probe. |

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Table 1 (*continued*)

| Parameter | Measures | Molecular probe(s) | Comments | Parameter | Measures | Molecular probe(s) | Comments |
|---|---|---|---|---|---|---|---|
| | $XYZ = XYZ_0$ | | | | | | Photophysical properties of dyes normalising between tetramethylsilane (0.00) to water (1.00) gives the $E_T^N$ scale. |
| | $+yY$ ("polarity") | | | | | | |
| | $+pP$ ("polarisability") | | | | | | |
| | $+eE$ (electophilicity) | | | | | | |
| | $+bB$ (nucleophilicity) | | | | | | |

concentrations of triethylphosphine oxide ($Et_3PO$) dissolved in the solvent.[30] The chemical shift is then extrapolated to infinite dilution. On the other hand, the Gutmann donor number (DN) is a measure of a solvent's Lewis basicity, derived from calorimetric (or computational[35]) measurements that provide the 1 : 1 enthalpies of interaction (in kJ $mol^{-1}$) between the solvent and excess $SbCl_5$ (both dissolved in 1,2-dichloroethane).[31] A solvent's Lewis acidity and basicity therefore cannot be directly compared *via* the Gutmann AN and DN, as they are defined on different unit scales. Similar problems are found for the $E_T$ and Kosower *Z*-values of other solvents.‡ For the $\alpha$, $\beta$ and $\pi^*$ scales in Kamlet–Taft's linear solvation energy relationships (LSERs), multiple probes are used for each formulation and whilst this does generalise the parameters, it does so statistically, by averaging across the different systems. These and other LSERs are further discussed in the ESI,† and Fig. S1 and S2.

Thus, despite each of these parameters having utility in a specific context – or perhaps because of it – there remains a limited fundamental understanding of the relationship between these distinct solvent parameters and specific non-covalent interactions that occur between solutes and solvents (*e.g.* electrostatics, exchange, induction, and dispersion). Essentially it is unclear what fundamental interactions these parameters are measuring. Herein we address this issue by investigating the fundamental intermolecular forces that occur between various ions and solvents using quantum chemical methods. The properties that quantitatively govern these physicochemical systems are then parameterised on the basis of the *ab initio* electrostatic descriptor þ ("sho"),[36] which has here been extended beyond its original purpose, to describe solvent environments. Finally, this new parameter is used to elucidate the molecular origins of existing empirical solvent parameters.

## Results and discussion

The Gutmann acceptor and donor numbers (a measure of solvent Lewis acidity and basicity respectively) have recently been proposed as good parameters for quantifying the solvent impact on observed specific ion and Hofmeister effects.[36,37] Strong correlations between these Lewis acidity or basicity scales with many other (seemingly unrelated) empirical solvent parameters (Table 1) have been shown repeatedly with few outliers (Fig. 1 and Fig. S2, ESI†), despite their fundamental differences.[25,38–40] This consistency indicates that each parameter is measuring some fundamental attribute of the solvent component of each probe-solvent interaction. Gutmann–Mayer's AN and Dimroth and Reichardt's $E_T(30)$ respectively claim to measure solvent electrophilicity and polarity, however Kamlet *et al.*[25] suggested that both parameters are linear combinations of these two measures. This conclusion arose since either AN or $E_T(30)$ parameters may be represented as

‡ Since pyridinium *N*-phenolbetaine is protonated in acidic solutions, no $E_T$-value can be found for solvents such as glacial acetic acid, whilst the 1-ethyl-4-methyoxycarbonylpyridinium iodide dye molecule used for Kosower *Z*-values has solubility issues in non-polar solvents.

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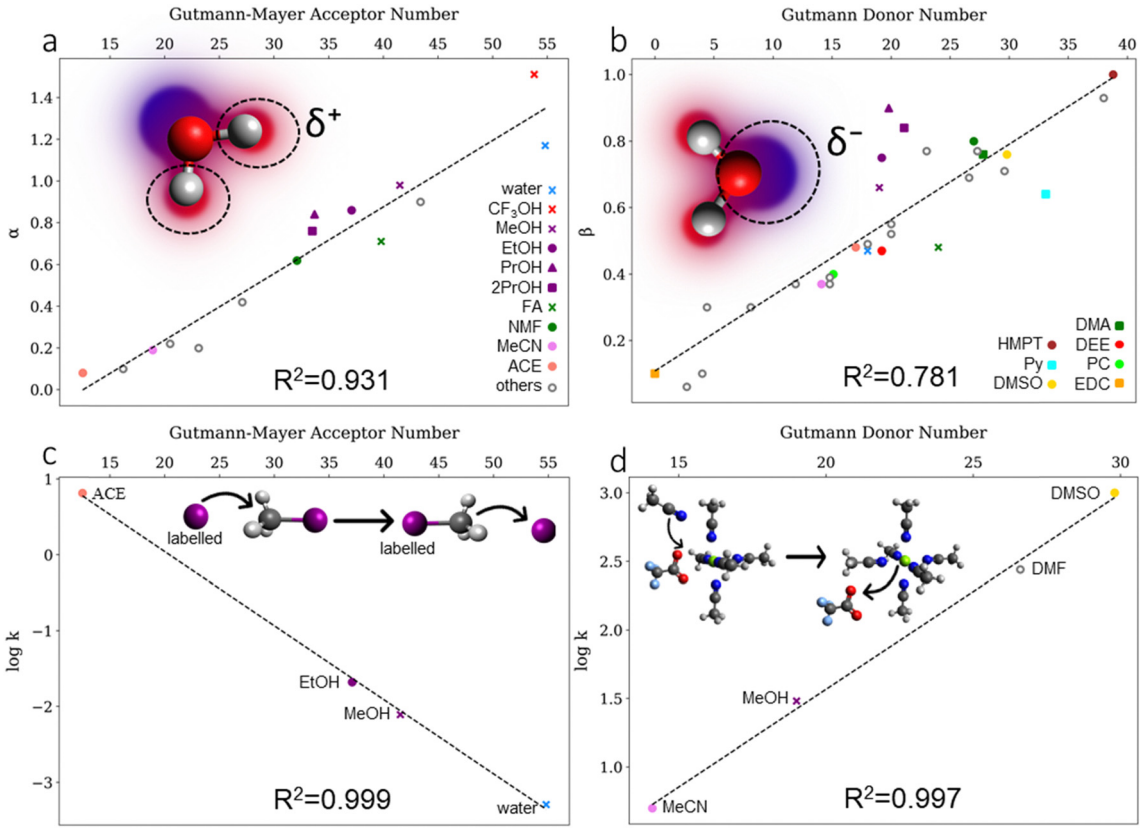



Fig. 1 Correlations between solvents' empirical parameters (Table 1) and various solvent properties. There is a general correlation between the Gutmann–Mayer acceptor number (AN) and the (a) Kamlet–Taft $\alpha$ values of Lewis acidity, and (b) the Gutmann donor number (DN) and the Kamlet–Taft $\beta$ values of Lewis basicity. These parameters correlate to many observed physicochemical phenomena, illustrated here using (c) rate constants of the 2nd order reaction between $CH_3I$ and $I^-$ and (d) the rate of solvent (*S*) substitution of trifluoroacetate ($TFA^-$) in $[NiS_5\ TFA]^+$. Data collated from ref. 41. Further examples provided in Fig. S2 (ESI†).

LSER, involving a combination of the electrophilicity and polarity terms.§ Fig. 1 therefore prompts two key questions regarding empirical solvent parameters. Firstly, which intermolecular forces (*i.e.* electrostatics, exchange, polarisation and dispersion) govern these consistent trends? Secondly, can these intermolecular forces be attributed to some fundamental property of the solvent molecules themselves that may be calculated?

We consider the first of these questions in two parts. Firstly, we investigate the intermolecular interactions of an ionic probe with a series of solvent molecules (Table 2 and Fig. S3–S6, ESI†) to reveal if any correlate with the Gutmann donor and acceptor numbers (Fig. S7 and Tables S2, S3, ESI†). Secondly, we make a direct comparison between the total interaction strength and each component energy (Fig. S8–S25, ESI†). Fig. 2(a) correlates cation–solvent interaction energies for a single solvent molecule in a surrounding solvent environment ($\Delta E$) with the solvent's Gutmann donor number. Fig. 2(b) correlates $\Delta E$ between anions and their surrounding solvent environment with the solvent's Gutmann–Mayer acceptor number. Fig. 2 indicates that the donor number is a 1 : 1 interaction ("solute") parameter, consistent with its calorimetric origin, whereas the acceptor number more closely relates to an entire solvation shell. Therefore, the donor number appears to relate to specific interactions while the acceptor number accords with bulk solvation properties, and therefore they have limited simultaneous use. For each of these interactions, the electrostatics correlates well with the total interaction (Fig. S8, S9 and Tables S2, S3, ESI†), indicating electrostatics has predictive potential in isolation of the other component energies. This is the case for both individual ion–solvent complexes calculated with gold-standard[42] SAPT2+3 and solvated ion–solvent complexes calculated with GKS-EDA.

By comparison, the other components of the total interaction energy are not consistently or significantly correlated with the total interaction energies. There are, however, other parameters that correlate well with these other terms (Fig. S26–S39, ESI†). A more complete discussion of these points is provided in the ESI† (Fig. S12–S39). Fig. 2 shows a marked correlation between $\Delta E$ and these empirical measures of acidity/basicity, with a high degree of correlation for each solute ion identity. These results are consistent with those of Hunter,[45] who demonstrated that the minimum and maximum electrostatic surface potentials of molecules correlate with their β (hydrogen-bond acceptor capacity) and α (hydrogen-bond donor capacity) respectively.

§ From ref. 25; $AN = 1.04 + 15.4(\pi^* - 0.08\delta) + 32.6\alpha$ and from ref. 25 and 108; $E_T(30) = 2.859(10.60 + 5.12(\pi^* - 0.23\delta) + 5.78\alpha)(10^3\ cm^{-1})$.

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Table 2 Solvents and their assigned acronyms, as well as the number of solvent molecules ($N$) used in GKS-EDA ion–solvent cluster calculation. The Gutmann donor number (DN),[43] Gutmann–Mayer acceptor number (AN),[44] relative permittivity ($\varepsilon_r$) at 25 °C,[44] dipole moment,[44] and $\text{þ}^{-/+}_{\text{solvent}}$ are also shown. Extended list of $\text{þ}^{-}_{\text{solvent}}$ and $\text{þ}^{+}_{\text{solvent}}$ is provided in the ESI (Table S1)

| Solvent | Acronym | $N$ | DN | AN | $\varepsilon_r$ | $\mu$ (D) | $\text{þ}^{-}_{\text{solvent}}$ ($\times 10^{-10}$ C m$^{-1}$) | $\text{þ}^{+}_{\text{solvent}}$ ($\times 10^{-10}$ C m$^{-1}$) |
|---|---|---|---|---|---|---|---|---|
| 1,2-Dichloroethane | EDC | 7 | 0 | 16.7 | 10.36 | 1.83 | −1.17 | 2.12 |
| 1-Propanol | PrOH | —[a] | 19.8 | 33.7 | 20.33 | 3.09 | −5.48 | 7.64 |
| 2-Propanol | 2PrOH | 8 | 21.1 | 33.5 | 19.92 | 1.66 | −5.72 | 7.51 |
| Acetone | ACE | 9 | 17 | 12.5 | 20.70 | 2.69 | −4.81 | 3.04 |
| Acetonitrile | MeCN | 15 | 14.1 | 18.9 | 35.94 | 3.92 | −4.10 | 3.23 |
| Ammonia | $NH_3$ | —[a] | 59 | — | 22.38 | 1.47 | −8.46 | 6.31 |
| Diethylether | DEE | 6 | 19.2 | 3 | 4.20 | 1.3 | −3.04 | 2.36 |
| Dimethylacetamide | DMA | 6 | 27.8 | 13.6 | 37.78 | 3.72 | −5.51 | 3.14 |
| Dimethylsulfoxide | DMSO | 9 | 29.8 | 19.3 | 46.68 | 4.06 | −5.98 | 3.40 |
| Ethanol | EtOH | 10 | 18.5 | 37.1 | 24.55 | 1.66 | −5.50 | 7.62 |
| Formamide | FA | 15 | 24 | 39.8 | 109.5 | 3.37 | −5.00 | 7.42 |
| Hexamethyl phosphoric triamide | HMPT | —[a] | 38.8 | 9.8 | 29.30 | 5.54 | −7.19 | 2.28 |
| Hexane | hexane | 5 | 0 | 0 | 1.88 | 0.08 | —[b] | 2.01 |
| Methanol | MeOH | 15 | 19 | 41.5 | 32.70 | 2.87 | −5.07 | 7.56 |
| *N*-Methylformamide | NMF | 9 | 27 | 32.1 | 182.4 | 3.86 | −4.90 | 5.51 |
| *N*-Methylacetamide | NMA | —[a] | — | — | 37.78 | — | −5.52 | 5.50 |
| Nitromethane | $MeNO_2$ | —[a] | 2.7 | 20.5 | 35.87 | 3.56 | −3.71 | 3.30 |
| Propylene carbonate | PC | 5 | 15.1 | 18.3 | 64.92 | 4.94 | −4.42[c] | 2.89 |
| Pyridine | Py | —[a] | 33.1 | 14.2 | 12.91 | 2.37 | −2.84 | 2.07 |
| Toluene | TOL | 5 | 0.1 | 0.3 | 2.38 | 0.43 | −1.15 | 2.41 |
| Water | water | 30 | 18 | 54.8 | 78.46 | 1.834 | −7.82 | 8.55 |

[a] Solvent only investigated *via* SAPT2+3 and DDEC6. [b] No surface partial negative charge for hexane. [c] This is a mixed value, based on the fact that PC interacts *via* multiple oxygens at once (Fig. S40, ESI).

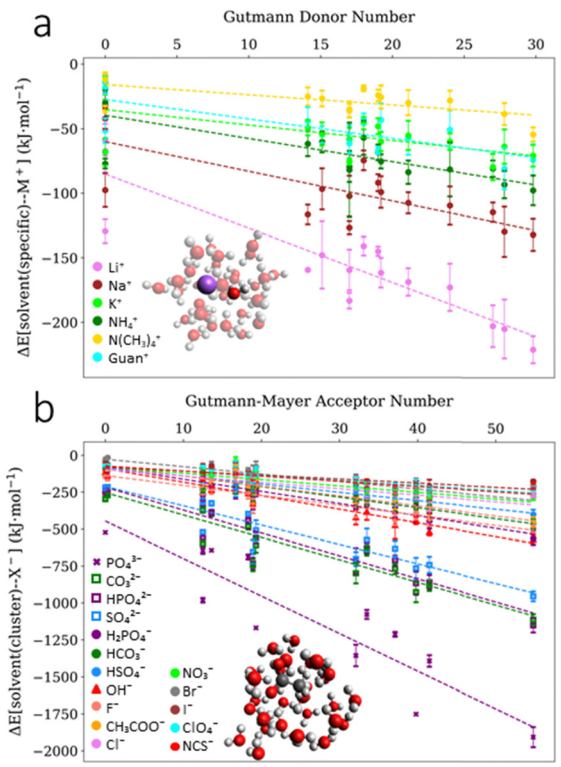



Fig. 2 Correlation between the solvent (a) acidity and (b) basicity, as measured *via* Gutmann donor/acceptor numbers, respectively, and the interaction between (a) cations and a specifically coordinated solvent molecule and (b) anions and their immediate solvent environment, as measured by GKS-EDA $\Delta E$ values. $R^2$ values for each ion–solvent trend presented in Tables S2 and S3 (ESI†).

The predominant correlation between electrostatic interactions and the overall solute–solvent interactions presented here (Fig. S8–S11, ESI†), and indeed solvent–solvent interactions more generally (Fig. S41–S46, ESI†), indicates that our recently proposed Coulombic þ parameter,[36] developed for predicting Hofmeister series and specific ion effects, might be extensible to describing solvent interactions more generally. If this hypothesis is correct, it would mean that empirical LSER and solvent parameters could be replaced with first-principles LSER that are self-consistent and based only on the fundamental interactions present in the system. To test this hypothesis, we develop the concept of a "solvent þ parameter" ($\text{þ}_{\text{solvent}}$). In its original formulation,[36] þ is an atom-centred approximation of the Coulombic interactions between an ion and its local environment, based on the effective charge ($q$) and radius $\left(\sqrt[3]{\langle r^3\rangle}\right)$ of the atom at the ion's interacting charge site,

$$U_{\text{E}} \propto = \text{þ}\frac{q}{\sqrt[3]{\langle r^3\rangle}}.$$

In most cases, the interacting charge site on an ion is known intuitively or can be determined *via* density functional or wavefunction theory methods, meaning that there is a more easily defined þ value for ions.[36] For molecular solvents, this is not necessarily the case, since charge will be more delocalised and the most charge-dense atoms are often shielded. Due to the greater structural complexity of solvents in comparison to the simple ions considered in the original formulation of þ, the concept of a solvent's þ value is more ambiguous. Nevertheless, Fig. 3(a) shows that the most meaningful $\text{þ}_{\text{solvent}}$ value generally coincides with the position of the positive and negative dipoles in the molecule, where the solvent would be expected to interact

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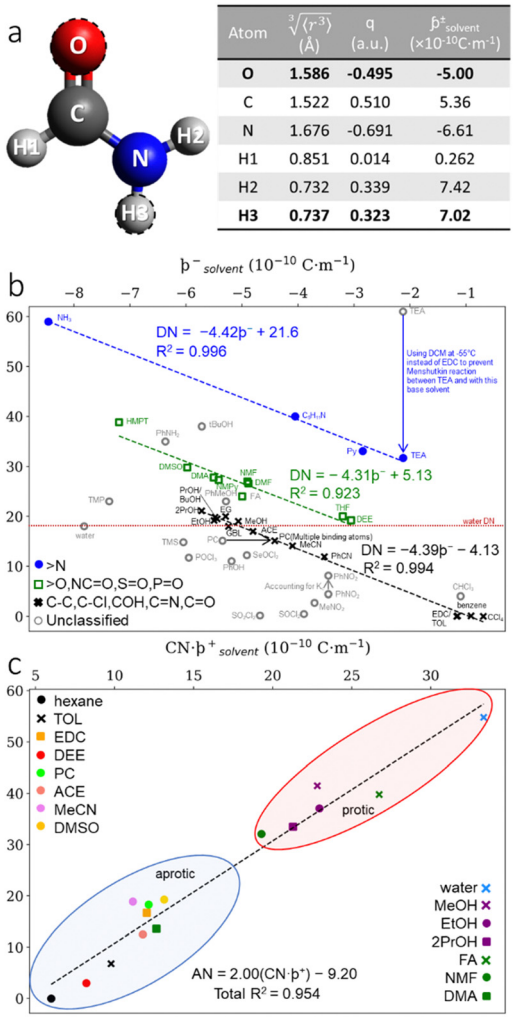



Fig. 3 (a) Formamide atomic radial moments, charges and $\flat^{\pm}_{solvent}$ values. Correlation between (b) the Gutmann donor number (DN) and $\flat^-_{solvent}$ values, and (c) the Gutmann–Mayer acceptor number (AN) and $\flat^+_{solvent}$ values. In (b), the dotted red line represents the donor number for water. The vertical blue arrow indicates a correction due to identifiable experimental errors in DN values. The horizontal arrow indicates multiple binding atoms need to be considered for $\flat^-_{solvent}$. CN in (c) indicates the coordination number of $Et_3PO$ at the oxygen atom (by hydrogen atoms) in each solvent, as calculated from DFTB-MD trajectories. Complete list of solvent acronyms and $\flat^{-/+}_{solvent}$ provided in Table S1 (ESI†).

with other molecules. For instance, in formamide, these are anticipated to reside on the oxygen and hydrogen (H3) atoms, respectively, which have the most accessible (large) negative and positive $\flat$ values ($\flat^- = -5.01 \times 10^{-10}$ and $\flat^+ = 7.13 \times 10^{-10}$ C m$^{-1}$, respectively). Note that despite the nitrogen and H2 atoms here having higher charge densities, they are not used since both are less chemically accessible (Fig. S30, S35 and S43, ESI†). $\flat^{\pm}_{solvent}$ data for the solvents investigated in this work are provided in Table S1 in ESI.†

The cross-correlations between the empirical solvent parameters listed in Fig. 1 and Fig. S2 (ESI†), and the correlations between these and the electrostatic energy of interaction (Fig. S47, ESI†), provides a strong indication that these empirical solvent parameters may ultimately arise from electrostatic interactions between the probe solute molecules and their solvent environments. Fig. S48(b) and (f) (ESI†) shows that these electrostatic interactions can be generally correlated with the $\flat^-_{solvent}$ or $\flat^+_{solvent}$ value of a given solvent, as appropriate (with some caveats; Fig. S49–S61, ESI†). It is therefore proposed that the Gutmann donor/acceptor numbers, and by extension many other empirical solvent parameters (Table 1 and Fig. S2, ESI†), are merely proxies for the electrostatic interactions associated with the negative or positive dipole of the solvent molecule respectively. Fig. 3 considers this possibility by examining the correlation between the empirical donor/acceptor numbers of solvents and their $\flat^-/\flat^+$ values. It is apparent from this figure that there is a strong and general correlation between solvent donor/acceptor numbers and the $\flat^-_{solvent}/\flat^+_{solvent}$ values, respectively. Interestingly, distinct correlations between $\flat^-_{solvent}$ and the Gutmann DN (Fig. 3(b)) intrinsically reflect the functional moiety that defines $\flat^-_{solvent}$ itself, with distinct linear relationships being observed for various functional groups, reflecting the trends observed in Fig. S1(a) (ESI†). Further, the trends for C═O bases (for example) appear additionally dependent on the adjoining functional groups (*e.g.* CC═O and NC═O appear to form different trends) and not just the donor atoms themselves. The functional group trend observed in Fig. 3(b) is reminiscent of those seen for ionic liquid anions, where N-donors had higher respective donor number than O-donors when compared to their DFT calculated interaction energy with a 1,3-dimethylimidazolium counterion.[46] Physically, the shift, or offset, in this general trend indicates that some other property or interaction not described by $\flat^-_{solvent}$ is systematically contributing to the measured DN. This appears especially relevant in molecules where bonds can be resonantly enhanced, such as pyridine or amides, likely due to the delocalisation of electrons (Tables S5 and S6, ESI†).[47] Nevertheless, the correlation between $\flat^-_{solvent}$ and the Gutmann DN (*i.e.* the gradient in Fig. 3(b)) is comparable irrespective of the functional group in question, which reflects that, fundamentally, each correlation arises from the same physical phenomenon irrespective of the functional group in question.

There are also many solvents that do not fall on one of the main general trends observed in Fig. 3(b). These solvents generally cannot be grouped into the functional group families represented in the main trends; for instance, many are chlorinated solvents. In this respect water, as a diprotic oxygen-based solvent, is unique; while $\flat^-_{solvent}$ is very sensitive to adjacent proticity/aliphatic functionalisation, the donor number is only minimally affected. For instance, water, EtOH and DEE with zero, one, and two ethyl groups respectively, have varied $\flat^-_{solvent}$ values of −7.82, −5.50, and $-3.05 \times 10^{-10}$ C m$^{-1}$, respectively, while their DN is fairly consistent, 18, 18.5, and 19.2 kcal mol$^{-1}$, respectively. In this respect, we hypothesise that nearby positive dipoles may weaken the observed interaction originating at the negative dipole (Fig. S50(c)–S54, ESI†). Laurence *et al.*[48] have highlighted that DN values themselves can also have inconsistencies resulting from experimental conditions. For instance,

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phosphoramides can induce ionisation of $SbCl_5$, while species such as $PhNO_2$ violate the assumption that it is completely complexed to $SbCl_5$ in solution. In the latter case, by accounting for the equilibrium constant ($K_C$) for $PhNO_2$–$SbCl_5$ formation, the $PhNO_2$ DN value increases from 4.4 to 8.1. There are also complications arising from the "inert" 1,2-dichloroethane (EDC) base solvent used to measure the Gutmann DN, which may react with triethylamine (TEA) *via* an $SbCl_5$-catalysed Menshutkin reaction. This is illustrated *via* the high donor number of 61 for TEA shown in Fig. 3(b), which is a clear outlier. However, using the value reported by Taft *et al.*,[49] which is based on alternative experiments at −55 °C in dichloromethane (DCM), this value decreases markedly to conform with the general trend observed in Fig. 3(b) for amines. It is therefore conceivable that, during the experimental determination of the DN, other complexations or reactions with the base solvent are common, with functional group-specific effects displacing the observed DN values from a single correlation with fundamental parameters such as $\beta^-_{solvent}$. Finally, solvents with multiple potential binding atoms, such as propylene carbonate (PC), appear to require each atom to be included in the definition of its $\beta^-_{solvent}$ value (Fig. S40, ESI†).

The Lewis acidity, as represented by Gutmann–Mayer AN differs from the DN in that it is not a 1 : 1 solvent : solute parameter (as shown in Fig. 2(b)). Instead, it is a measure of the total solute–solvent interaction, meaning that the solute–solvent coordination number needs to be considered (Fig. S62, ESI†). We also note that $\beta^+_{solvent}$ is always located on a hydrogen in the solvent set investigated here, rather than differing elements as is the case for the DN functional group family trends. Accordingly, for the $\beta^+_{solvent}$/acceptor number correlation (shown in Fig. 3(c)) the $\beta^+_{solvent}$ values are scaled by the (hydrogen) coordination number of the $Et_3PO$ oxygen atom in each respective solvent (Table S7, ESI†). The coordination numbers were obtained from quantum chemical molecular dynamics simulations. Therefore, knowing the coordination number of a solute in any given solvent becomes an important consideration, especially if attempting to use the two parameters together (something ill-advised anyway for DN and AN, as they measure different things, and seemingly different numbers of molecules) – a likely necessity for ionic solutes where simultaneous interactions at the solvent molecules' positive and negative dipoles is potentially important. In this regard, the recent IonSolvR database[50,51] may be used for coordination number analysis of a wider array of solvents and ions. Previous work relating to $E_T(30)$ trends suggested that alcohols can be treated as binary mixtures of their OH moiety and the less polar alkyl chain.[52] With increasing carbon chain length the effective OH concentration tends to zero. This would apply to AN measurements where the probability of specific OH interactions occurring (effectively the CN) will be depleted with increasing chain length, therefore reducing the deshielding effect measured *via* NMR for that stronger interacting functional group. However, non-primary alcohols were shown to deviate from these chain-length trends with $E_T(30)$,[52] yet this is not the case for normalising $\beta^+_{solvent}$ by the CN, which inherently accounts for alterations in solvent properties (*e.g.* solvent packing) that underly those deviations. Also, notably, while the binary mixture treatment to account for chain length appears to result in linear trends for pure solvents, non-linear trends are observed for actual binary mixtures.[52,53] This is hypothesized to be due to the preferential solvation by one solvent over the other.

The $\beta^\pm_{solvent}$ parameters are site-specific, so would likely require multiple parameters for complex poly-functional group solvents. While relevant for describing micro-solvation effects, where local intermolecular interactions are dominant, they may struggle to describe macroscopic solvation phenomena related to the overall stabilisation of a solute.[54] However, it is conceivable that adaptions of the Lorentz–Lorent or Claussius–Mossoti equations, or indeed a Taylor series expansion of the electrostatic energy,[55] which relate microscopic and macroscopic properties, may be generated to relate $\beta^\pm_{solvent}$ to properties such as the relative permittivity (Fig. S63–S67, ESI†). Further work to investigate the effects of concentration, temperature and mixtures, in conjunction with the molecular properties is still required to fully understand how the colligative, additive and constitutional properties of solvents[56] may be fully encompassed by a fundamental $\beta^\pm_{solvent}$ analysis. Nonetheless, given $\beta^-_{solvent}$ and $\beta^+_{solvent}$ are both fundamental solvent parameters derived *via* the same methodology, a dual usage of these parameters, and indeed a deeper understanding of how solvents interact with various species, is now possible. This is especially important for considering the modulation effects of a solvent in a unified manner with ubiquitous specific ion effects in a unified manner.[36,37,57]

## Conclusions

Over the last 75 years, a multitude of empirical solvent parameters, including the Kamlet–Taft parameters and Gutmann donor and acceptor numbers, have been proposed to examine and quantify a range of specific physicochemical phenomena. While these phenomena are often disparate, many experimentally determined solvent parameters have in common the fact that they rely on a specific molecular probe, or property. They are therefore empirical descriptors of these phenomena, yet it is conceivable that they are simply proxies for some fundamental interaction(s) between solutes and their solvent environments. On the basis of this hypothesis, here we have presented results demonstrating that the fundamental ion/solvent–solvent intermolecular interaction is predominantly an electrostatic interaction in nature, and that the $\beta^\pm_{solvent}$ parameters are, in effect, a first principles and self-consistent solvent parameter capable of replacing traditional empirical parameters (such as those representing Lewis acidity and basicity). $\beta^\pm_{solvent}$ provides a new, first-principles tool for predicting and understanding the physicochemical behaviour of solvents and chemical systems that have previously relied on parameters such as those listed in Table 1. Further, we believe that the $\beta^\pm_{solvent}$ will inform the development of improved solvation models, which often depend on dielectric solvent properties, when paired with higher order electrostatic terms

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(dispersion, induction, *etc.*) and make them extensible to more complex solvent environments such as ionic liquids[58,59] and deep eutectic solvents.[60]

## Computational methods

Randomised initial ion–single solvent molecule geometries were created *via* the packmol[61] package, from optimised solvent and ion structures. These ion–solvent complexes were optimised with Truhlar's M06-2X[62] exchange correlation functional in conjunction with an aug-cc-pVDZ[63] basis set, using the Gaussian16 program.[64] Symmetry-adapted perturbation theory[42,65–67] including a complete 3rd-order description (*i.e.* SAPT2+3[68–70]) with an aug-cc-pVDZ[63] basis set was used. SAPT2+3 was implemented in the Psi4 package,[71] using initial guess orbitals from the superposition of atomic densities,[72,73] to calculate the electrostatic, exchange, repulsion, induction, dispersion and total interaction energy contributions at M06-2X/aug-cc-pVDZ – optimised geometries. For ions containing heavier elements, pseudopotentials and more diffuse functions were used, *i.e.*, aug-cc-pVDZ-PP[74–77] for $I^-$ or aug-cc-pVDZ-X2C[78–81] for $K^+$. For determining the partial charge, atomic radii and as well as $C_6$ dispersion coefficients (using the Manz-Chen-Cole-Limas-Fiszbein (MCLF) method[82]) of all isolated ion and solvent molecules, a density-derived electrostatic and chemical (DDEC6) charge distribution analysis[83] was performed using MP2[84]/aug-cc-pVTZ for solvent molecules. This charge distribution method appears to be largely independent of the functional and basis set choice (Fig. S68, ESI†).

Ion–solvent cluster interactions were modelled *via* individual ions solvated in *N*-solvent clusters (the value of *N* varied for each solvent due to computational costs, whilst maintaining the first solvation shell, see Table 2). All clusters were generated using packmol[85] and optimised using 3rd order density functional tight binding (DFTB) theory[86] including D3(BJ) dispersion[87,88] and the 3ob-3-1 parameter set.[89–92] GKS-EDA(sol)[93] calculations were then performed at the M06-2X[62]/cc-pVDZ[94] level of theory, including a PCM solvent model to approximate the surrounding bulk solvent environment, and a counterpoise correction[95] to account for basis set superposition error. All GKS-EDA(sol) calculations were performed using GAMESS,[96] and all DFTB calculations were performed using DFTB+.[97] To ensure a statistical description of the ion–solvent interaction this procedure was repeated 10 times for each ion–solvent combination, using randomised starting geometries. Interaction energies between the ion and the solvent environment, and the components of these energies, were calculated *via* the schemes presented in Fig. S37 in ESI.†

$Et_3PO$-solvent coordination numbers (CN) were determined *via* molecular dynamics (MD) data generated using the DFTB+ software package (v. 19.1).[97] Initial geometries for these trajectories consisted of random ensembles of 64¶ solvent molecules, about the $Et_3PO$ probe molecule, generated using the packmol package.[61] MD simulations were performed using 3rd order density functional tight binding (DFTB3),[86] which was computed on-the-fly at each timestep using the 3ob-3-1 parameter[90–92] set. Grimme's two-body D3 dispersion[87,88] with Becke-Johnson[98,99] dampening (*i.e.* D3(BJ)) was included throughout all simulations. Charge mixing was configured with the Broyden method.[100,101] All MD trajectories were performed using constant volume & temperature dynamics (*i.e.* NVT ensemble) *via* a Nosé[102]–Hoover[103] chain[104,105] (NHC) thermostat (chain-length = 3) set to 300 K with a coupling constant of 1000 $cm^{-1}$. Solvent densities were held at the experimental solvent density of the pure solvent throughout all simulations.[106] Periodic boundary conditions (PBC) were enforced on all trajectories (cubic unit cell), with charges handled *via* particle mesh Ewald summation.[107] All MD trajectories were iterated using a timestep of 1 fs, with coordinates and relevant information recorded every 10 fs. The performance of this method for reproducing experimental and *ab initio* radial distribution functions is discussed and investigated in our previous work.[51]

¶ 32 solvent molecules were used for hexane.

## Author contributions

K. P. G. and A. J. P. were responsible for the conceptualisation of the project. K. P. G. performed and analysed all calculations and collated all the data. K. P. G. and A. J. P. drafted original version of the manuscript. K. P. G. prepared the ESI.† All authors contributed to reviewing and editing the final manuscript.

## Conflicts of interest

The authors declare no competing interests.

## Acknowledgements

A. J. P., G. B. W., V. S. J. C. and E. J. W. acknowledge Australian Research Council funding (ARC DP190100788, DP230102030 and LE170100032 (INTERSECT)). K. P. G. acknowledges an Australian Government Research Training Program (RTP) Scholarship. This research was undertaken with the assistance of resources provided at the NCI National Facility systems at the Australian National University, through the National Computational Merit Allocation Scheme supported by the Australian Government. The authors would like to thank Gareth Elliott, Dr Edwin Johnson and Dr Hayden Robertson for valuable discussions.

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